

CS254 – Network Security

Sample research projects (do not redistribute)

Research project – 50%

- A list will be given or you can choose your own
 - ▣ 2 students form a group (individual is also fine)
 - ▣ Grading based on contribution percentage
- General goals
 - ▣ Analyze a system to identify weaknesses
 - ▣ Propose a new defense / **Re-implement** or adapt a known work
 - ▣ Aim for a publishable workshop paper (or something you can brag about)

Presentation logistics

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- Transition takes a lot of time
 - ▣ Next group/presenter standing in line
 - ▣ Order specified on the web sheet
- Only VGA/HDMI connectors are available

Sample research project 1

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- Analyze the network traffic of a smartphone app
 - ▣ Identify risks of using unencrypted connections
 - ▣ What can an attacker do?
- Goals
 - ▣ Understand risks of using plaintext in network communication

Sample research project 2

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- Analyze the network traffic of a networked game (can be a simple one)
 - ▣ Reverse engineer its protocol
 - ▣ Build a bot client
- Goals
 - ▣ Understand the challenges of defending against automated bot clients

Sample research project 3

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- Study ingress filtering
 - ▣ How widely is ingress filtering done?
 - ▣ Where do they happen?
 - ▣ Does home modem do ingress filtering?
- Goals
 - ▣ Measurement
 - ▣ (optional) Hack the home modem

Sample research project 4

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- Survey and test out the presence of deployed defenses against attacks mentioned in the paper
 - ▣ ARP, ICMP redirect, RIP etc.
- Goals
 - ▣ Learn the difference between what happens in theory and what happens in practice

Sample research project 5

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- Analyze and develop strategies of encrypting TCP sequence number
 - ▣ IPSec
 - ▣ SPDY
 - ▣ QUIC
- Goals
 - ▣ Understand tradeoffs
 - ▣ Develop new strategies

Sample research project 6

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- Understand the censorship mechanism
 - ▣ How does the latest censorship firewall work?
 - How does it detect sensitive connections?
 - How does it stop the communication?
- Goal
 - ▣ Picture the innerworkings of censorship firewall
 - ▣ Develop strategies to evade censorship
 - E.g., Can we use protocols such as QUIC (which are not yet recognized by most firewalls) to evade censorship?

Sample research project 7

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- DNS cache poisoning attack
 - ▣ SAD DNS is a prominent example that can revive DNS cache poisoning attack through side channels
- Goal
 - ▣ Investigate the feasibility of other side channels (one concrete candidate is IPID) that can potentially lead to the same attack

Sample research project 8

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- DNS protocol implementation flaws
 - ▣ DNS is often offered as a system service
 - ▣ Global state may be influenced maliciously
- One concrete problem we discovered on Windows
 - ▣ RTT estimates in protocols may be polluted -> DoS
- Goal
 - ▣ Understand how the RTT estimates are implemented through source code or disassembled binaries
 - ▣ Identify what other global state may be influenced

Sample research project 9

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- IPv6 protocol security
 - ▣ We are in a transition from IPv4 to IPv6
 - ▣ Understanding the security of IPv6 is critical
- Goal
 - ▣ Using existing scanning tools to test vulnerabilities of various IPv6 protocol implementations (Windows, Linux, FreeBSD, Mac OS)
 - ▣ Understand how current tools can achieve various attacks, e.g., parasite6 that can achieve MITM in LAN, dos-new-ip6 that can DoS new IPv6 devices.
- Reference
 - ▣ <https://github.com/vanhauser-thc/thc-ipv6>

Sample research project 10

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- IP Hopping and its application of defeating IP-based checks, e.g., voting
 - ▣ If you own a cell phone, you can potentially change your 4G/3G/2G IP address forcefully
 - ▣ This allows you to bypass many web services that perform IP-based rate limit (e.g., each IP is allowed to vote once)
- Research goal
 - ▣ Analyze the speed at which IP hopping can be done on cell phones
 - ▣ Survey the set of major web services that are vulnerable

Others

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- Implement, re-implement, or adapt a published attack
 - ▣ Vulnerabilities published on the Internet (CVE database), but no publicly available exploits
 - ▣ Some research papers described an attack but no code is provided
 - ▣ Sometimes PoC (with source code) is published but it only targets one specific environment
- Benefits
 - ▣ You get a concrete understanding of how to execute a fully functional attack

Sample re-implementation projects

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- ❑ [Spring4shell vulnerability: https://sysdig.com/blog/cve-2022-22965-spring-core-spring4shell/](https://sysdig.com/blog/cve-2022-22965-spring-core-spring4shell/)
- ❑ [Log4j vulnerability: https://logging.apache.org/log4j/2.x/security.html](https://logging.apache.org/log4j/2.x/security.html)
- ❑ [ShadowMove: A Stealthy Lateral Movement Strategy, USENIX Security 2020](#)
- ❑ [IMP4GT: IMPersonation Attacks in 4G NeTworks, NDSS 2020](#)
- ❑ [Practical Traffic Analysis Attacks on Secure Messaging Applications, NDSS 2020](#)
- ❑ [Fragmentation Considered Poisonous](#)
- ❑ SSL/TLS attacks: BEAST, BREACH, CRIME, TIME, etc.
- ❑ Side channels in web applications. [Side-Channel Leaks in Web Applications: a Reality Today, a Challenge Tomorrow](#)
- ❑ Evading a firewall such as Bro. [Insertion, Evasion, and Denial Of Service: Eluding Network Intrusion Detection](#)